Jamuna S. Murthy

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RESEARCH INTERESTS

Vision–language models and scalable AI systems — models that ground language in images, video, and audio, and the efficient architectures and infrastructure that run them from edge devices to the cloud. Current focus: **theory of controllable image generation**, treating controllability as a property of structured latent geometry, and measuring what generated images lose in meaning.

EDUCATION

M.Tech. in Computer Software Engineering, Ramaiah Institute of Technology, Bengaluru — 2015 – 2017
Autonomous, affiliated to Visvesvaraya Technological University · GPA 9.30/10
Thesis: *Distributed Framework for Real-Time Twitter Sentiment Analysis and Visualization (TwitSenti)* — Best Master's Thesis Award

B.E. in Computer Science and Engineering, Dr. Ambedkar Institute of Technology, Bengaluru — 2011 – 2015
Autonomous, affiliated to Visvesvaraya Technological University

PUBLICATIONS

Under Submission

1. **Jamuna S. Murthy**, Amin Karimi Monsefi, Rajiv Ramnath. Controlla: Learning Controllability via Graph-Constrained Latent Geometry. *arXiv:2605.16603*, May 2026. UNDER SUBMISSION · A*
2. **Jamuna S. Murthy**, Amin Karimi Monsefi, Rajiv Ramnath. KathaTrace: Diagnosing Semantic Trajectory Collapse in Generated Visual Narratives. *arXiv:2607.01312*, July 2026. UNDER SUBMISSION · A*

Journal Articles

3. **Jamuna S. Murthy**, G. M. Siddesh. Multimodal Humor Detection in Videos via Domain-Specific Representations and Deep Hybrid Fusion Networks. *Soft Computing* (Springer Nature), 2026. ACCEPTED SEP 2026
4. **Jamuna S. Murthy**, G. M. Siddesh. Multimedia Video Analytics using Deep Hybrid Fusion Algorithm. *Multimedia Tools and Applications* (Springer), 2025.
5. **Jamuna S. Murthy**, G. M. Siddesh. A Smart Video Analytical Framework for Sarcasm Detection using Adaptive Fusion and SarcasNet-99. *The Visual Computer* (Springer), 2024.
6. **Jamuna S. Murthy**, G. M. Siddesh, Wen-Cheng Lai, et al. ObjectDetect: A Real-Time Object Detection Framework for Advanced Driver Assistant Systems Using YOLOv5. *Wireless Communications and Mobile Computing* (Wiley), 2022.
7. **Jamuna S. Murthy**, G. M. Siddesh, K. G. Srinivasa. TwitSenti: A Real-Time Twitter Sentiment Analysis and Visualization Framework. *Journal of Information & Knowledge Management*, 2019.
8. **Jamuna S. Murthy**, G. M. Siddesh, K. G. Srinivasa. A Real-Time Twitter Trend Analysis and Visualization Framework. *International Journal on Semantic Web and Information Systems*, 2019.

Conference Papers

9. Abhay Bhandarkar, **Jamuna S. Murthy**, Amogh Kalasapura. CAMMEL-ViT: Context-Aware Memory with Lightweight Vision Transformers for Medical Image Segmentation. *Medical Image Understanding and Analysis (MIUA)*, University College Dublin, 2026.
10. **Jamuna S. Murthy**, Dhanashekar Kandaswamy, Wen-Cheng Lai. Towards Secure Video Surveillance: A Few-Shot Spatiotemporal Perception Transformer for Unseen Behavioral Anomalies. *IEEE International Conference on Advanced Visual and Signal-Based Systems (AVSS)*, Tainan, 2025. CORE A

PATENTS AND FUNDING

Surgeon's Eye: Intelligent Video Analytical Framework for Robotic and Laparoscopic Surgery 2025
Patent application No. 202541008338 · Jamuna S. Murthy, Dhanashekar Kandaswamy, G. M. Siddesh

FitQua: Smart Water Bottle for AI-Based Health Tracking — DST-funded project 2021 – 2022
Thermal control with sensor fusion (temperature, load, activity) for AI-driven hydration monitoring.

ACADEMIC APPOINTMENTS

Assistant Professor, Department of CSE, Ramaiah Institute of Technology, India 2022 – present

- Lab affiliation: Center for Imaging Technologies.
- Faculty sponsor, RIT ACM Student Chapter (2023 – present).

Assistant Professor, Department of ISE, BNM Institute of Technology, India 2020 – 2022

- Head, Google Cloud Center of Excellence.

Assistant Professor, Department of CSE, PES University, India 2018 – 2020

- Lab affiliation: Centre for Data Science & Applied Machine Learning (CDSAML).

Research Assistant, Department of Computational and Data Sciences, Indian Institute of Science, Bengaluru 2017 – 2018

INDUSTRY RESEARCH EXPERIENCE

AI Researcher, Verity Labs Inc. (AI-ACC-Stack), California (remote) Apr 2025 – Sep 2025

- Modular AI backend stack: retrieval-augmented generation, LLM integration, document ingestion, tool-calling agents, structured querying, and deployable AI services.

Faculty Mentor, Samsung PRISM (Samsung R&D), India Jun 2024 – May 2025

- Expression Correction in Photos: GAN-based affective computing models for emotion-aware image enhancement.
- NextGen AI Datastore for Video Retrieval (ongoing): vector database for multimodal video retrieval with vision–language embeddings.

Intern, Amazon Jun 2016 – Jun 2017

ACADEMIC SERVICE

Reviewer	Machine learning: NeurIPS (2026) · AAAI (2025, 2026, 2027) · AISTATS (2026, 2027) Computer vision: ICCV (2025) · ECCV (2026) Multimedia: ACM Multimedia (2025, 2026, 2027) · ACM Multimedia Asia (2025, 2026)
Membership	Member, Association for Computing Machinery (ACM), since 2020 Lifetime Member, Indian Society for Technical Education (ISTE)

HONORS AND AWARDS

Young Researcher Award, STEM Research Society Nov 2024
Best Master's Thesis Award, Ramaiah Institute of Technology May 2017
TEQIP-funded Graduate Teaching Assistantship, Ramaiah Institute of Technology 2016 – 2017

TEACHING

Ramaiah Institute of Technology (undergraduate): Artificial Intelligence and Machine Learning (CS52) · Deep Learning · Cloud Computing and Big Data (CS62)